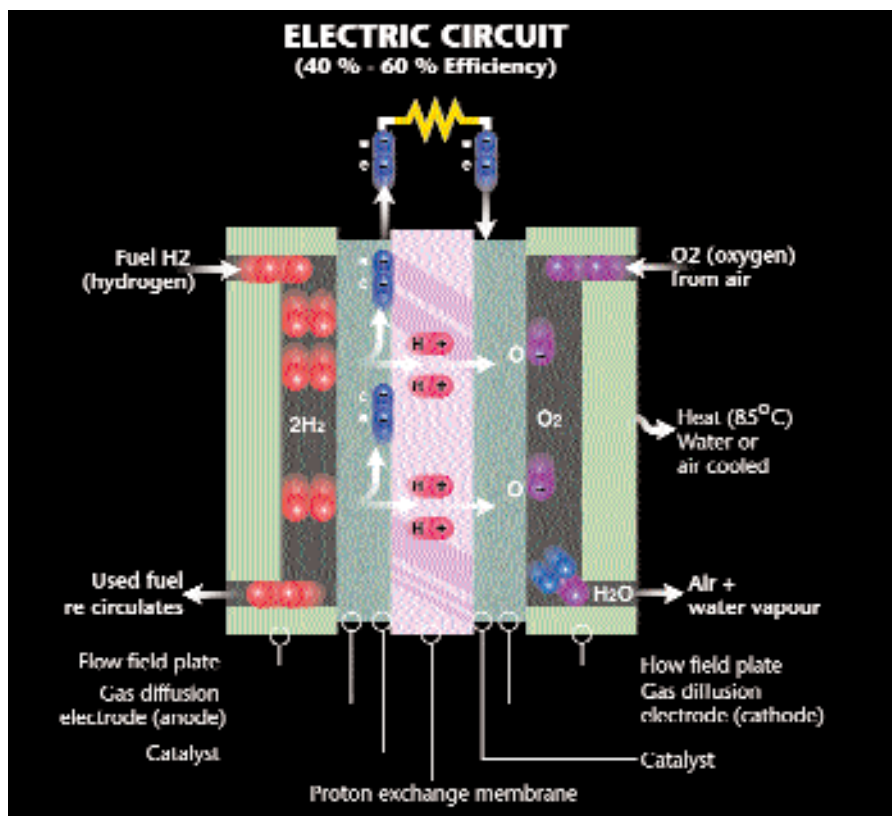




Jargon Buster

Electrolyte: This is a chemical compound which, when fused or dissolved in certain solvents, usually water, will conduct an electric current. All electrolytes in the fused state, or in solution, give rise to ions which conduct the electric current.

Energy density: This is the ratio of the amount of energy in the cell, to the weight or volume of the cell. It can be measured in watt-hours per pound, or watt-hours per cubic inch.

Cycle Life: This defines the number of times a battery may be charged and discharged. Every time a battery is charged and discharged it uses one cycle.

Capacity: The amount of energy a battery can store is its capacity. Measured in amp hours or milliamp hours, this is the length of time the battery can supply the required voltage.

maH: The milliamp-hour rating indicates the number of milliamps the cell can produce per hour. This means that a 1,000 milliamp-hour battery could produce 10 milliamps for 100 hours, or 100 milliamps for 10 hours, or (theoretically) 1,000 milliamps for 1 hour, or even 2,000 milliamps for 30 minutes.

Memory Effect: Nickel-based batteries in particular, tend to 'remember' their usual discharge point. For example, if you constantly recharge your battery after 50 per cent consumption, after a period of time, the remaining 50 per cent of your battery chemicals become ineffective. This causes the effective battery life to become only 50 per cent of what it was.

Self Discharge: A percentage of capacity that the battery will discharge for a given time, while the battery is not in use.

being tested with relatively heavy Li-ion packs which last up to 8 hours. MFCs could drastically reduce the weight carried, and give much longer run times with the added benefit of almost instant recharging.

Japan's NEC recently unveiled a laptop running on a methanol fuel cell instead of a standard battery. The device runs for about 5 hours in continuous operation, and is expected to rise to as much as 40 hours in 2 years time. Even when the cell does run down, it will only take a quick refill of fuel to get you up and running, instead of the long recharges associated with typical batteries. NEC's Japanese rival Toshiba has also announced a similar fuel cell powered laptop. Keep an eye out for both in 2004.

A company called Smart Fuel Cell has also developed a usable micro fuel cell for laptops and expects to produce 100,000 units by 2004. The prototype holds 120 ml of methanol and generates about 150 Wh—enough to power a 15 W notebook computer for 10 hours. Recharging simply means changing the cartridge.

NTT DoCoMo, who provide third generation (3G) mobile services in Japan, also have plans to commercialise fuel cell powered 3G phones, as early as 2004. With large colour screens, high-speed Internet access and video conferencing, current 3G

phones are plagued by low battery life. It is hoped that the new phone will at least match the battery life provided by their current 2G models. Users can carry around a cigarette lighter type container, that will recharge the phone's fuel cell.

Fuel cells unfortunately may be cumbersome to deal with, just like non-reusable batteries. Consumers don't like the idea of having to go out and buy replacement fuel packs. Smart Fuel Cell's laptop cell comes with tamper-proof methanol cartridges that snap into the device. But, this raises the issue of recurring cost to the user, where he has to go out and buy fuel. Methanol cartridges should cost about \$5 (Rs 230). The only other alternative is to feed in fuel manually, which isn't a tempting prospect either.

Availability of the right fuel will be another issue for initial adopters. It will probably be a while before you can go to a convenience store and ask for a refill can of methanol or hydrogen. The solution to this problem does not lie in making hydrogen widely available, but rather, in making fuel cells that use readily available fuel—beer, for example, or for that matter almost any type of alcohol served locally. Researchers at St Louis University have carried out development on a technology called the Biofuel cell. A Biofuel cell uses

enzymes to catalyse the reaction. This means you could theoretically take any type of alcohol and pour it into your device—and it would be cheap too. One of the researchers claims that once the system is fully optimised, one to three drops of alcohol could power your cell phone for a month. Fuel cells also require air (for the oxygen) and do emit minute amounts of carbon dioxide and water vapour. A micro fuel cell maker claims that the quantities of water vapour emitted are similar to the amount you have on your palm right now and should not be of concern.

One way or another, there is no stopping fuel cells. US President George Bush has asked congress for \$1.2 billion to fund